Class 12 Mathematics

Chapter 12: Linear Programming

Practice Set B: step-wise solutions

10 questions **27** marks **3** levels

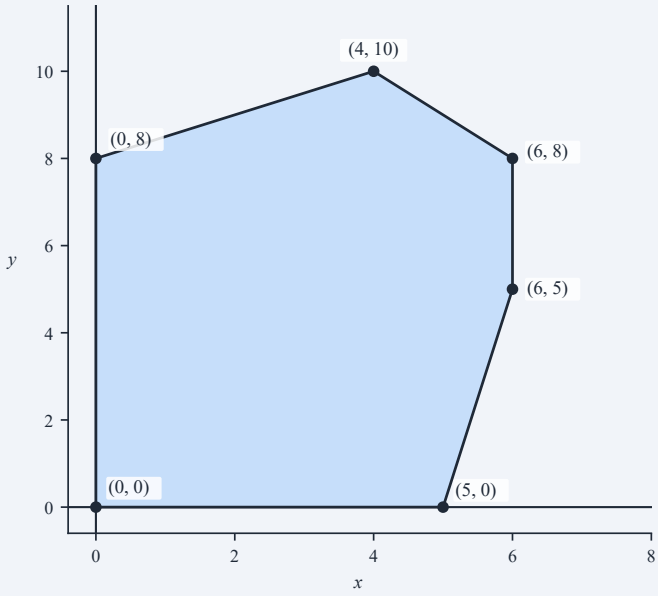
Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

4 marks

1 **MCQ** NCERT Exemplar — 1 mark

The feasible region for an LPP is shown in the figure. Let $Z = 3x - 4y$ be the objective function. (Maximum value of Z + Minimum value of Z) is equal to:



(A) 13 **(B)** 1 **(C)** -13 **(D)** -17

Formula / idea: On a bounded feasible region, the maximum and minimum of Z occur at corner points.

1 Evaluate $Z = 3x - 4y$ at the first three corner points.
$(0, 0):\ Z = 0$
$(5, 0):\ Z = 15$
$(6, 5):\ Z = 18 - 20 = -2$

2 Evaluate Z at the other three corner points.
$(6, 8):\ Z = 18 - 32 = -14$
$(4, 10):\ Z = 12 - 40 = -28$
$(0, 8):\ Z = -32$

3 The region is bounded, so read off the largest and smallest values.
$Z_{\max} = 15$ at $(5, 0)$, $Z_{\min} = -32$ at $(0, 8)$

4 Add the two values. ✓ 1
$15 + (-32) = -17$

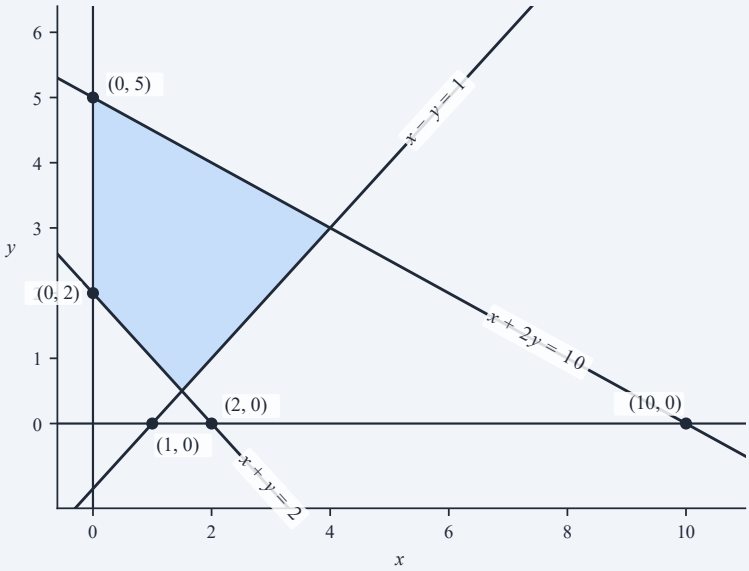
5 Pick the matching option.
Option (D)

Answer **(D)** -17

Watch out: Taking $Z_{\min} = 0$ at the origin: the $-4y$ term makes Z negative at the upper corners.

2 MCQ CBSE SQP 2023-24 1 mark

The feasible region corresponding to the linear constraints of a Linear Programming Problem is given below. Which of the following is not a constraint to the given Linear Programming Problem?



(A) $x + y \geq 2$ **(B)** $x + 2y \leq 10$ **(C)** $x - y \geq 1$ **(D)** $x - y \leq 1$

Formula / idea: Every point of the feasible region satisfies every constraint of the LPP.

1 Take a point inside the shaded region.
$P(1, 3)$

2 Test P in options (A) and (B).
$1 + 3 = 4 \geq 2$ ✓, $1 + 6 = 7 \leq 10$ ✓

3 Test P in $x - y$: it satisfies (D) but not (C). ✓ 1
$1 - 3 = -2 \leq 1$ ✓, $-2 \geq 1$ is false

4 So the region lies on the side $x - y \leq 1$; $x - y \geq 1$ is not a constraint.
Option (C)

Answer **(C)** $x - y \geq 1$

Watch out: Testing with the origin: $(0, 0)$ is not in the region, so it cannot decide which side is shaded.

3 MCQ Board-pattern 1 mark

For a linear programming problem with objective function $Z = ax + by$, which of the following statements is **false**?

(A) If the feasible region is unbounded, a maximum or a minimum value of Z may not exist.

(B) A corner point of a feasible region is a point in the region which is the intersection of two boundary lines.

(C) **If the origin is a corner point of the feasible region, then the minimum value of Z is always 0.**

(D) Points within and on the boundary of the feasible region represent feasible solutions.

Formula / idea: $Z = ax + by$ can take negative values when a or b is negative.

1 (A) is the textbook remark on unbounded regions: true.

2 (B) and (D) are the textbook definitions of a corner point and of feasible solutions: true.

3 Test (C): take $Z = -x$ on the region with corners $(0, 0)$, $(1, 0)$, $(0, 1)$. ✓ 1
$Z(0, 0) = 0, \quad Z(1, 0) = -1, \quad Z(0, 1) = 0 \Rightarrow Z_{\min} = -1 \neq 0$

4 So (C) is false.
Option (C)

Answer **(C) If the origin is a corner point of the feasible region, then the minimum value of Z is always 0.**

Watch out: Assuming Z is never negative: that holds only when $a, b \geq 0$.

4 **Assertion-Reason** Board-pattern 1 mark

Assertion (A): The linear programming problem: Maximise $Z = x + y$ subject to $x - y \leq -1$, $-x + y \leq 0$, $x \geq 0$, $y \geq 0$ has no optimal solution. **Reason (R):** The feasible region of this linear programming problem is unbounded.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) **Assertion (A) is true, but Reason (R) is false.**

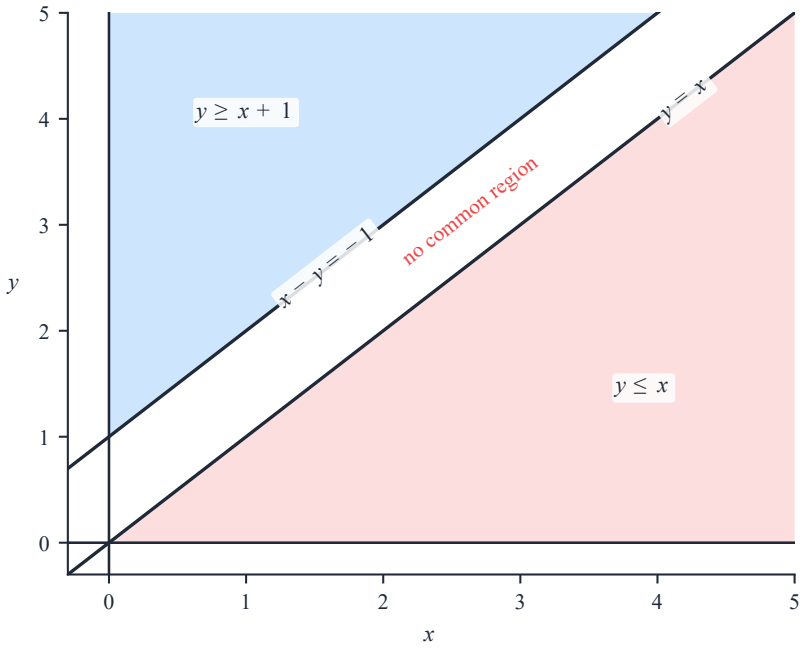
(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: If no point satisfies all the constraints, there is no feasible region and no optimal solution.

1 Rewrite the two constraints.

$x - y \leq -1 \Leftrightarrow y \geq x + 1, \qquad -x + y \leq 0 \Leftrightarrow y \leq x$

2 No point has $y \geq x + 1$ and $y \leq x$ together: the half planes do not meet.



3 No feasible point, so no optimal solution.

A is true

4 The feasible region is empty, not unbounded.

R is false

5 R is false, so it cannot explain A.

6 Hence the correct option is (C). ✓ 1

Option (C)

Answer **(C) Assertion (A) is true, but Reason (R) is false.**

Watch out: Choosing (A): the region does not extend indefinitely, it does not exist at all.

Level 2

9 marks

5 **Short answer** NCERT Exemplar 3 marks

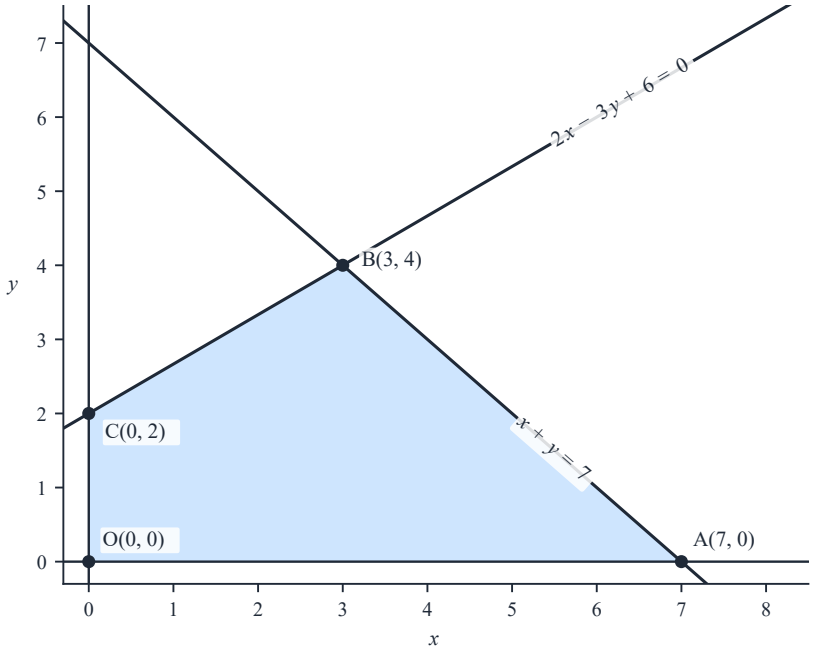
Solve the following linear programming problem graphically: Minimise $Z = 13x - 15y$ subject to the constraints $x + y \leq 7, 2x - 3y + 6 \geq 0, x \geq 0, y \geq 0$.

Formula / idea: Corner Point Method: on a bounded feasible region, the minimum of Z occurs at a corner point.

1 Draw the constraint lines.

$$x + y = 7: \ (7, 0), \ (0, 7)$$
$$2x - 3y + 6 = 0: \ (0, 2), \ (3, 4)$$

2 Origin test: $0 \leq 7$ and $6 \geq 0$ are true, so shade towards O. Bounded region $OABC$. ✓ 1



3 Corners on the axes. ✓ ½

$O(0, 0), \quad A(7, 0), \quad C(0, 2)$

4 Corner B: put $y = 7 - x$ in $2x - 3y = -6$. ✓ ½

$2x - 21 + 3x = -6 \ \Rightarrow \ x = 3, \ y = 4 \ \Rightarrow \ B(3, 4)$

5 Evaluate $Z = 13x - 15y$ at each corner point. ✓ ½

$O(0,0):\ Z = 0$

$A(7,0):\ Z = 91$

$B(3,4):\ Z = 39 - 60 = -21$

$C(0,2):\ Z = -30$

6 The region is bounded, so the least corner value is the minimum. ✓ ½

$Z_{\min} = -30$ at $(0,2)$

Answer **Minimum $Z = -30$ at $x = 0, y = 2$.**

Watch out: Answering $Z_{\min} = 0$ at the origin: the negative coefficient of y gives smaller values.

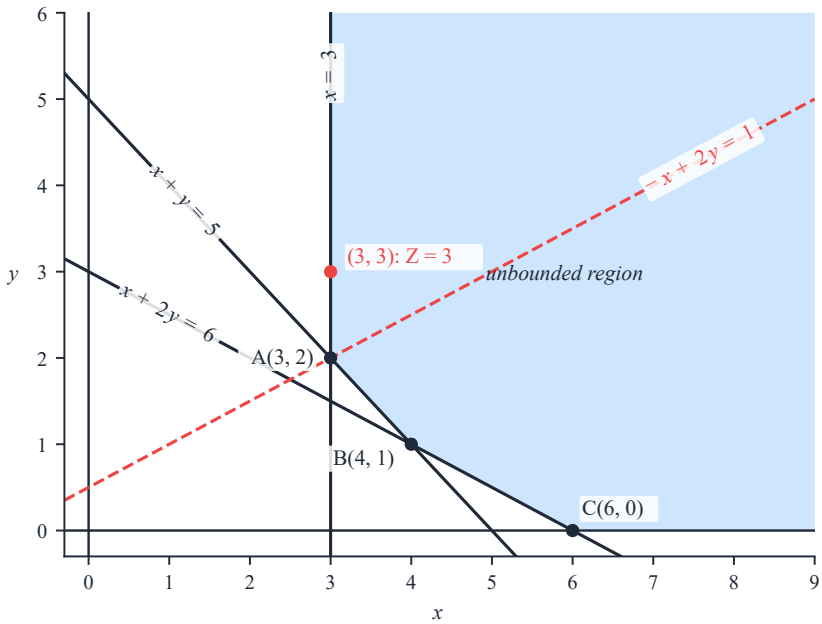
6 **Short answer** CBSE SQP 2023-24 3 marks

Solve the following Linear Programming Problem graphically: Maximise $Z = -x + 2y$ subject to the constraints $x \geq 3, x + y \geq 5, x + 2y \geq 6, y \geq 0$.

Formula / idea: Unbounded region: the largest corner value M is the maximum only if $Z > M$ has no common point with it.

1 Draw $x = 3$, $x + y = 5$ through $(5,0), (0,5)$ and $x + 2y = 6$ through $(6,0), (0,3)$.

2 All are $\geq$: shade away from the origin. The feasible region is unbounded. ✓ 1



3 Corners: $x = 3$ with $x + y = 5$; $x + y = 5$ with $x + 2y = 6$; $x + 2y = 6$ with $y = 0$. ✓ ½

$A(3,2), \quad B(4,1), \quad C(6,0)$

4 Evaluate $Z = -x + 2y$ at the corner points. ✓ ½

$Z(A) = -3 + 4 = 1, \quad Z(B) = -4 + 2 = -2, \quad Z(C) = -6$

5 Largest corner value is $M = 1$. Region is unbounded, so check $-x + 2y > 1$.

6 Point $(3, 3)$ is feasible and lies in this half plane. ✓ ½

$3 \geq 3,\ \ 6 \geq 5,\ \ 9 \geq 6; \qquad Z(3, 3) = 3 > 1$

7 So $-x + 2y > 1$ meets the feasible region; hence Z has no maximum value. ✓ ½

Answer Z **has no maximum value (the feasible region is unbounded and** $-x + 2y > 1$ **meets it).**

Watch out: Writing $Z_{\max} = 1$ at $(3, 2)$: on an unbounded region, test the half plane $-x + 2y > 1$ first.

7 **Short answer** NCERT Exemplar 3 marks

Solve the following linear programming problem graphically: Maximise $Z = x + y$ subject to the constraints $x + 4y \leq 8$, $2x + 3y \leq 12$, $3x + y \leq 9$, $x \geq 0$, $y \geq 0$.

Formula / idea: Corner Point Method: on a bounded feasible region, the maximum of Z occurs at a corner point.

1 Draw the constraint lines using their intercepts.

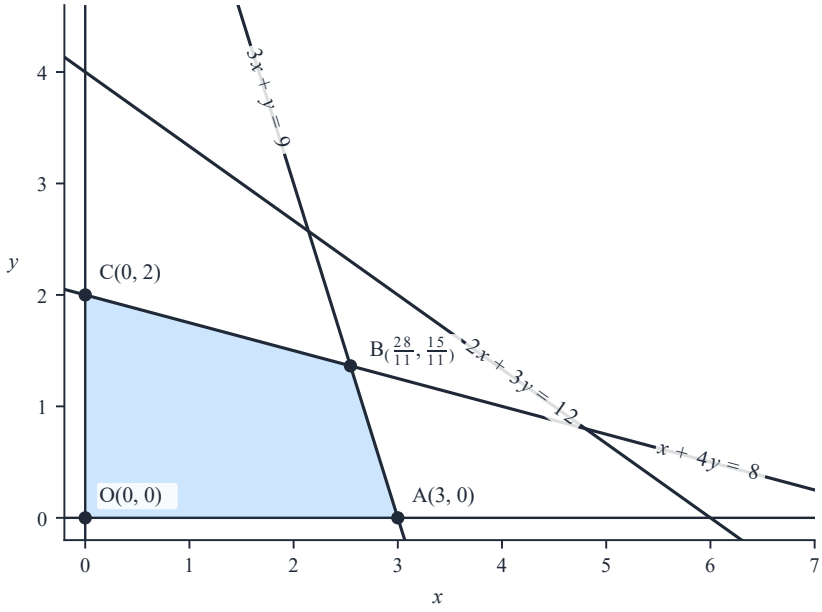
$x + 4y = 8:\ (8, 0),\ (0, 2)$

$2x + 3y = 12:\ (6, 0),\ (0, 4)$

$3x + y = 9:\ (3, 0),\ (0, 9)$

2 All are $\leq$: shade towards O. Bounded region $OABC$; $2x + 3y = 12$ gives no edge. ✓ 1

Every point of $OABC$ already satisfies $2x + 3y \leq 12$.



3 Corners on the axes. ✓ ½

$O(0, 0), \quad A(3, 0), \quad C(0, 2)$

4 Corner B: put $y = 9 - 3x$ in $x + 4y = 8$.

$x + 36 - 12x = 8 \Rightarrow x = \frac{28}{11},\ y = \frac{15}{11}$

✓ ½

5 Evaluate $Z = x + y$ at each corner point.

$O:\ Z = 0$

$A(3, 0):\ Z = 3$

$B\left(\frac{28}{11}, \frac{15}{11}\right):\ Z = \frac{43}{11}$

$C(0, 2):\ Z = 2$

✓ ½

6 The largest corner value is the maximum.

$Z_{\max} = \frac{43}{11}$ at $\left(\frac{28}{11}, \frac{15}{11}\right)$

✓ ½

Answer **Maximum** $Z = \frac{43}{11}$ **at** $x = \frac{28}{11}, y = \frac{15}{11}$.

Watch out: Taking $2x + 3y = 12$ as an edge: its meeting points with the other lines lie outside the region.

Level 3

14 marks

8 **Long answer** NCERT Exemplar 5 marks

Solve the following linear programming problem graphically: Maximise and minimise $Z = 3x - 4y$ subject to the constraints $x - 2y \leq 0, -3x + y \leq 4, x - y \leq 6, x \geq 0, y \geq 0$.

Formula / idea: Unbounded region: test $Z > M$ for the maximum and $Z < m$ for the minimum.

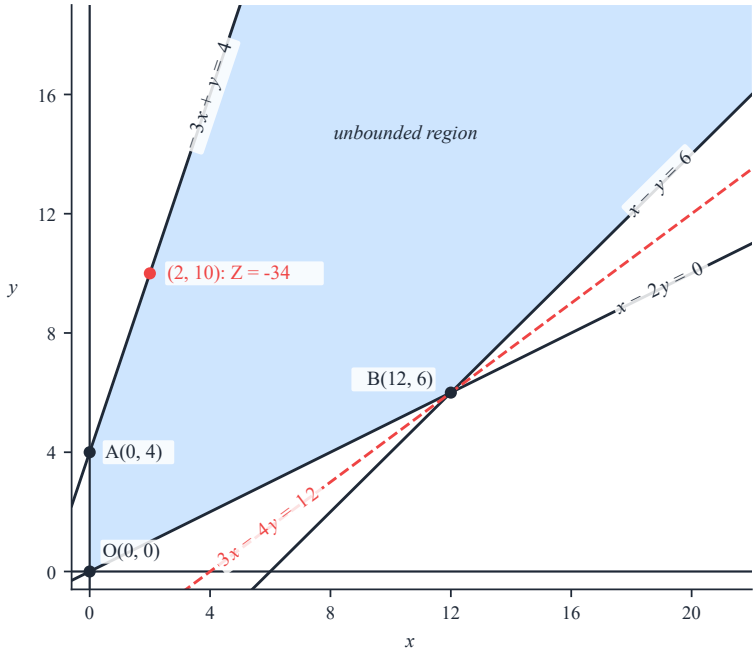
1 Draw the constraint lines.

$x - 2y = 0:\ (0, 0),\ (12, 6)$

$-3x + y = 4:\ (0, 4),\ (2, 10)$

$x - y = 6:\ (6, 0),\ (12, 6)$

2 Shade above $x - 2y = 0$, below $-3x + y = 4$ and above $x - y = 6$. Unbounded region. ✓ 1½
Test $(0, 1)$ for $x - 2y \leq 0$; $(0, 0)$ for the others.



3 Corners on the y-axis. ✓ ½
$O(0, 0), \qquad A(0, 4)$

4 Corner B: put $x = 2y$ in $x - y = 6$. ✓ ½
$2y - y = 6 \Rightarrow y = 6,\ x = 12 \Rightarrow B(12, 6)$

5 Evaluate $Z = 3x - 4y$ at the corner points. ✓ ½
$Z(O) = 0, \qquad Z(A) = -16, \qquad Z(B) = 36 - 24 = 12$

6 Largest corner value is $M = 12$. Region is unbounded, so draw $3x - 4y = 12$.

7 The open half plane $3x - 4y > 12$ has no point in common with the region. ✓ ½
On the edge $y = x - 6$, $Z = 24 - x \leq 12$.

8 Hence the maximum exists. ✓ ½
$Z_{\max} = 12$ at $(12, 6)$

9 Smallest value $m = -16$. Point $(2, 10)$ is feasible and gives $Z < -16$. ✓ ½
$2 - 20 \leq 0, \quad -6 + 10 \leq 4, \quad 2 - 10 \leq 6; \qquad Z(2, 10) = -34$

10 So $3x - 4y < -16$ meets the feasible region; hence Z has no minimum value. ✓ ½

Answer **Maximum $Z = 12$ at $(12, 6)$; Z has no minimum value.**

Watch out: Writing $Z_{\min} = -16$ at $(0, 4)$: the region is unbounded and Z keeps decreasing upwards.

9 **Long answer** NCERT Exemplar 5 marks

A company manufactures two types of sweaters: type A and type B. It costs ₹360 to make a type A sweater and ₹120 to make a type B sweater. The company can make at most 300 sweaters and spend at most ₹72000 a day. The number of sweaters of type B cannot exceed the number of sweaters of type A by more than 100. The company makes a profit of ₹200 for each sweater of type A and ₹120 for every sweater of type B. Formulate this problem as an LPP and solve it graphically to find how many sweaters of each type the company should make in a day to get the maximum profit. Also find the maximum profit.

Formula / idea: Formulate (variables, objective function, constraints), then use the Corner Point Method.

1 Let x = type A sweaters and y = type B sweaters made per day.

2 Write the profit to be maximised. ✓ ½

$$Z = 200x + 120y$$

3 Cost constraint; divide by 120. ✓ ½

$$360x + 120y \le 72000 \;\Rightarrow\; 3x + y \le 600$$

4 Number of sweaters constraint. ✓ ½

$$x + y \le 300$$

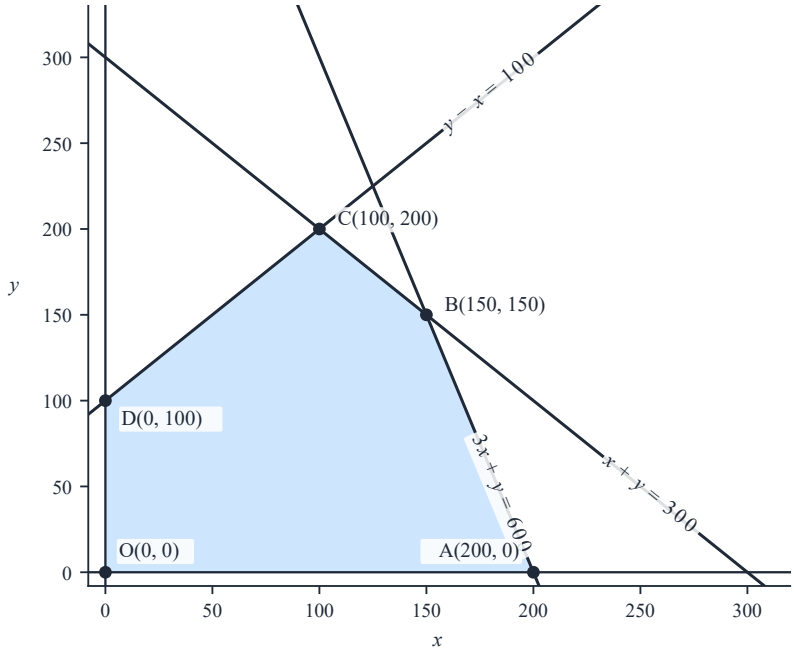
5 Type B exceeds type A by at most 100; numbers are non-negative. ✓ ½

$$y - x \le 100, \qquad x \ge 0,\ y \ge 0$$

6 Draw the constraint lines.

$$3x + y = 600:\ (200, 0),\ (0, 600)$$
$$x + y = 300:\ (300, 0),\ (0, 300)$$
$$y - x = 100:\ (0, 100),\ (100, 200)$$

7 Origin test is true for all three: shade towards O. Bounded region $OABCD$. ✓ 1



8 Corners on the axes: $3x + y = 600$ meets the x-axis, $y - x = 100$ meets the y-axis.
$O(0,0), \quad A(200,0), \quad D(0,100)$

9 Corner B: subtract $x + y = 300$ from $3x + y = 600$. ✓ ½
$2x = 300 \Rightarrow x = 150,\ y = 150 \Rightarrow B(150,150)$

10 Corner C: put $y = x + 100$ in $x + y = 300$. ✓ ½
$2x = 200 \Rightarrow x = 100,\ y = 200 \Rightarrow C(100,200)$

11 Evaluate $Z = 200x + 120y$ at O, A and B.
$O(0,0):\ Z = 0$
$A(200,0):\ Z = 40000$
$B(150,150):\ Z = 30000 + 18000 = 48000$

12 Evaluate Z at C and D. ✓ ½
$C(100,200):\ Z = 20000 + 24000 = 44000$
$D(0,100):\ Z = 12000$

13 Largest corner value is the maximum.
$Z_{\max} = 48000$ at $B(150,150)$

14 Make 150 sweaters of each type a day; maximum profit ₹48000. ✓ ½

Answer **150 sweaters of type A and 150 of type B a day; maximum profit ₹48000.**

Watch out: Writing $x - y \le 100$: it is type B that may exceed type A, so $y - x \le 100$.

10 **Case study** NCERT Exemplar 4 marks

Case Study: A company makes three models A, B and C of calculators at two factories I and II. It has orders for at least 6400 calculators of model A, 4000 of model B and 4800 of model C. Every day, factory I makes 50 calculators of model A, 50 of model B and 30 of model C, while factory II makes 40 of model A, 20 of model B and 40 of model C. It costs ₹12000 a day to operate factory I and ₹15000 a day to operate factory II. Let factory I operate for x days and factory II for y days.

(i) Write the objective function and the constraints of the LPP in simplest form. 1 mark

(ii) Find the corner points of the feasible region. 1 mark

(iii) Find the minimum operating cost, and justify why the minimum exists even though the feasible region is unbounded. 2 marks

Formula / idea: Unbounded region: the smallest corner value m is the minimum only if $Z < m$ has no common point with it.

(i) 1 mark

1 Operating cost to be minimised. ✓ ½
$Z = 12000x + 15000y$

2 Orders for models A, B, C, each divided to its simplest form. ✓ ½

$50x + 40y \geq 6400 \Rightarrow 5x + 4y \geq 640$

$50x + 20y \geq 4000 \Rightarrow 5x + 2y \geq 400$

$30x + 40y \geq 4800 \Rightarrow 3x + 4y \geq 480$

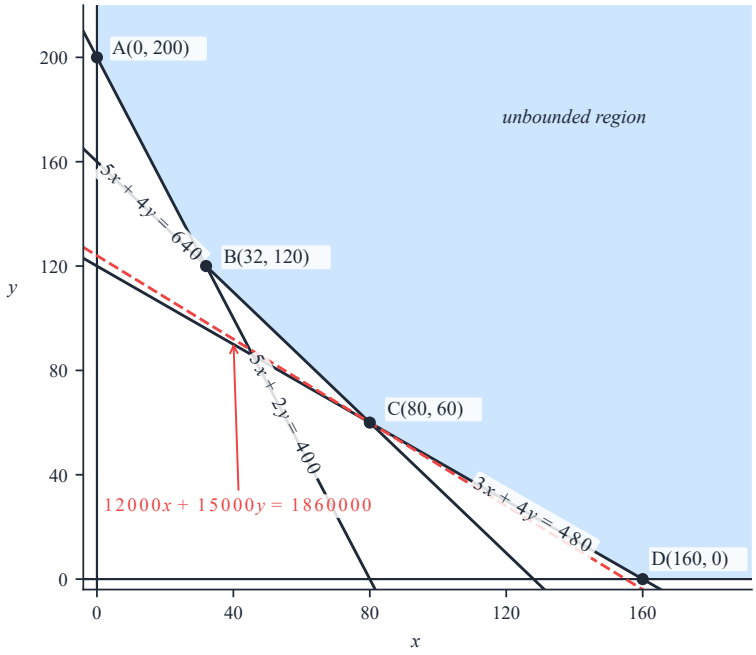
$x \geq 0, \quad y \geq 0$

Answer (i) **Minimise** $Z = 12000x + 15000y$ **subject to** $5x + 4y \geq 640$, $5x + 2y \geq 400$, $3x + 4y \geq 480$, $x, y \geq 0$

(ii)

1 mark

1 Draw the three lines; all are $\geq$, so shade away from the origin.



2 Axis corners: $5x + 2y = 400$ meets the y-axis, $3x + 4y = 480$ meets the x-axis. ✓ ½

$A(0, 200), \qquad D(160, 0)$

3 Subtract $5x + 2y = 400$ from $5x + 4y = 640$; subtract $3x + 4y = 480$ from $5x + 4y = 640$. ✓ ½

$2y = 240 \Rightarrow B(32, 120); \qquad 2x = 160 \Rightarrow C(80, 60)$

Answer (ii) $A(0, 200)$, $B(32, 120)$, $C(80, 60)$, $D(160, 0)$

(iii)

2 marks

1 Evaluate $Z = 12000x + 15000y$ at each corner point. ✓ ½

$A(0, 200): \; Z = 30{,}00{,}000$

$B(32, 120): \; Z = 3{,}84{,}000 + 18{,}00{,}000 = 21{,}84{,}000$

$C(80, 60): \; Z = 9{,}60{,}000 + 9{,}00{,}000 = 18{,}60{,}000$

$D(160, 0): \; Z = 19{,}20{,}000$

2 The smallest corner value is $18{,}60{,}000$, at $C(80, 60)$. ✓ ½

3	Region is unbounded, so draw the open half plane $Z < 18{,}60{,}000$. $12000x + 15000y < 1860000 \Leftrightarrow 4x + 5y < 620$	✓ ½
4	It has no point in common with the feasible region, so this is the minimum. $\text{Minimum cost} = ₹\,18{,}60{,}000 \text{ at } x = 80,\ y = 60$ The line $4x + 5y = 620$ touches the region only at C.	✓ ½

Answer (iii) **Minimum cost ₹18,60,000, with factory I for 80 days and factory II for 60 days; it exists because $4x + 5y < 620$ has no point in common with the feasible region.**

Watch out: Taking $(0, 160)$ as a corner: it fails $5x + 2y \geq 400$, so the y-axis corner is $(0, 200)$.